

SIMATS ENGINEERING

ASSESSMENT TOOL 1

COURSE NAME: ARTIFICIAL INTELLIGENCE COURSE CODE: CSA17

ASSESSMENT TOOL 1 – SUMMARY REPORT

#	Problem	Method	Result
1	Water Jug	Breadth-First Search	3G filled first; 8 moves; 4 gal isolated in 5-gal jug
2	Mars Rover	PEAS + agent analysis	Model-Based Utility-Based Agent recommended
3	8-Queens	Backtracking Search	Columns 1-8 -> Rows 8,4,1,3,6,2,7,5 (valid)
4	OLA Cab Booking	Utility-based agent (scoring)	Sedan selected, utility = 19.00
5	Logistics Network	Uniform Cost Search	S -> B -> D -> G, min cost = 7

1. Water Jug Problem

State represented as (5-gallon, 3-gallon). Starting from (0,0), BFS explored the reachable states level by level and returned the shortest sequence of fill/empty/pour moves. This run fills the 3-gallon jug first, reaching the target in 8 moves — the 5-gallon jug ends with exactly 4 gallons.

2. Mars Rover

Answered as a PEAS-style analysis: percepts come from cameras, spectrometers and environmental sensors; the environment is partially observable, dynamic, sequential, continuous and single-agent; actions include driving, sampling and transmitting data; performance is measured by scientific yield, coverage, communication reliability, power efficiency and safety. A Model-Based Utility-Based Agent architecture was recommended since the rover must weigh competing options under partial observability.

3. 8-Queens Problem

Formulated as a state-space search (initial state, state space, actions, goal test, path cost) and solved with backtracking, which places one queen per column, trying lower rows first, and abandons a partial placement the moment it becomes unsafe. The search returned the valid arrangement Column→Row = (1,8) (2,4) (3,1) (4,3) (5,6) (6,2) (7,7) (8,5).

4. OLA Cab Booking

Identified as a Utility-Based Problem-Solving Agent, since the booking decision must trade off comfort, fare and waiting time rather than just check reachability. Pseudocode queries nearby drivers per cab type, scores each by a weighted utility function, and books the highest-scoring option. In the worked example, Sedan was selected (utility = 19.00) over the higher-comfort Prime option because Prime's longer wait time outweighed its comfort advantage under the chosen weights.

5. Logistics Network

Modeled as a weighted graph over warehouses S, A, B, C, D, G and solved with Uniform Cost Search, which always expands the lowest cumulative-cost frontier node via a priority queue. The search expanded S, B, A, D, C, G in that order and returned S→B→D→G at a minimum total cost of 7, which is cheaper than the more direct-looking S→A→D→G (cost 11) or S→B→G (cost 10).

All computational results above were produced by U1_A1_Python_Code.py and match the worked solutions in U1_A1_Solution.pdf.